

SmartDoc OCR

Agentic Image-to-Word Converter

From Static Tool → Intelligent Agent

Professional Practices in IT (PPIT) — Final Project Phase 2 • FAST-NUCES, Islamabad — BSAI

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Live Demo · GitHub · Deployed on Streamlit Cloud

Phase 1 Was Solid — But It Was Just a Tool

Phase 1 Achievements

- Working Streamlit app: Image → Word converter
- 6-stage preprocessing: upscale → grayscale → denoise → CLAHE → adaptive threshold → deskew
- Heading detection (#, ALL CAPS), bullet detection (•, -, 1.), L/C/R alignment, bold via font height
- Stack: Python · Tesseract 5.4 · OpenCV 4.13 · python-docx · Streamlit
-  Deployed live on Streamlit Cloud
-  Verified with 12-point test suite

Where Phase 1 Falls Short

- **Static logic** — same pipeline for every image (good or bad)
- **No autonomy** — user must pick all settings manually
- **No memory** — every session starts from scratch
- **No explanation** — black-box formatting decisions
- **No learning** — never improves over time

 Phase 1 = Tool. Phase 2 = Agent. We had to evolve.

Building Software is a Professional Responsibility



USERS

Their documents pass through our app. Wrong OCR = wasted time & lost trust.



SOCIETY

Tools democratize digitization. Poor quality widens the digital divide.



DATA

Users trust us with personal/academic content. We must protect it.

Morals (Personal)

“I feel uncomfortable storing user data” — subjective, varies by person

Examples: privacy instincts, individual beliefs about openness

Ethics (Professional)

Codified rules that apply to all professionals (e.g., GDPR mindset)

Example actions: don't store user data; follow licensing & attribution

❏ Our SmartDoc OCR design follows **professional responsibility**: quality & privacy prioritized; transparency enabled.



How SmartDoc OCR Honors the ACM & IEEE Code of Ethics

Principle	Where We Applied It
Public Interest — ACM 1.1	Free, open-source tool — helps students & professionals globally
High Quality — ACM 2.1	6-stage preprocessing + 12-point verification suite + multi-PSM OCR
Privacy — ACM 1.6	Session-only processing, zero persistence, no tracking
Honesty — ACM 1.3	Confidence scores displayed; limitations (handwriting) disclosed
Competence — IEEE 6	Tesseract 5.4, OpenCV 4.13, EasyOCR — audited, proven libraries

Built-in Safeguards

- **Code:** Modular utils.py shared Phase 1 & 2
- **Security:** File-type validation (jpg/png/bmp/tiff), 100 MB cap, HTTPS
- **Safety:** Confidence warnings (<50% red, <70% yellow, ≥70% green); downloadable audit log; human override

We don't just follow the code — we engineered our system to enforce it.

Ethics in Action — A Real Decision From Our Project

Ethical Lenses Applied

- **Utilitarianism:** Free deployment + 6-stage preprocessing benefits all users — adopted
- **Deontology:** Duty to accuracy & privacy — non-negotiable — adopted
- **Human-Centered Design:** User overrides, visible confidence, controlled downloads — embedded

Decision: Speed vs Quality in Preprocessing

Option	Time	Accuracy	Verdict
Direct Tesseract on raw image	~0.5 s	40–50%	Rejected — unreliable
6-stage preprocessing pipeline	~3–5 s	65–75% (measured)	Adopted — prioritizes user accuracy

📄 Ethical principle: "The interests of the user outweigh our convenience as developers."

From Ethical Theory → Industry-Standard Practice

01

Identify Issue

Users upload sensitive academic/personal documents

02

Analyze Stakeholders

Users (privacy), Developers (training needs), Regulators (PECA 2016, GDPR)

03

Evaluate Alternatives

A) Store silently ❌ · B) Ask consent ⌚ · C) Session-only ✅

04

Justified Decision

Zero persistence — session-only processing declared in UI

Student Project vs Industry Software House

Practice	Our Project	Industry Standard
Version control	Git + GitHub	Git + CI/CD
Code review	Peer review	Formal PR + tests
Architecture	Modular utils.py	Microservices
Reproducibility	requirements.txt + packages.txt	Docker + IaC
AI ethics	Confidence shown, limits disclosed	Formal review board

Legal Compliance — Pakistani & Global Standards

Data Protection

In-memory processing only — no DB, no content logs. PECA 2016 §§3 & 9 considered. GDPR-minded data minimization.

Intellectual Property (IPR)

All major libraries Apache 2.0 / MIT — attributed. Our code = original, MIT licensed, public on GitHub.

IPR Violations Avoided

No proprietary APIs without license, no unattributed copy-paste, no scraping paid datasets.

Library License Status

Tesseract OCR	Apache 2.0 — Attributed
OpenCV	Apache 2.0 — Attributed
python-docx	MIT — Attributed
Streamlit	Apache 2.0 — Attributed
EasyOCR	Apache 2.0 — Attributed

Risks, Contracts & How We Mitigate Them

Risk Matrix

Risk	Likelihood · Mitigation
Data theft of uploads	Low · Session-only, HTTPS, no storage
User uploads copyrighted material	Medium · T&C disclaimers; user responsibility
Malicious file disguised as image	Low · File-type validation + 100 MB cap
OCR misread → wrong document	Medium · Confidence scores + warnings + user review
Code theft	Low · MIT license enforces attribution

Implicit Terms of Service

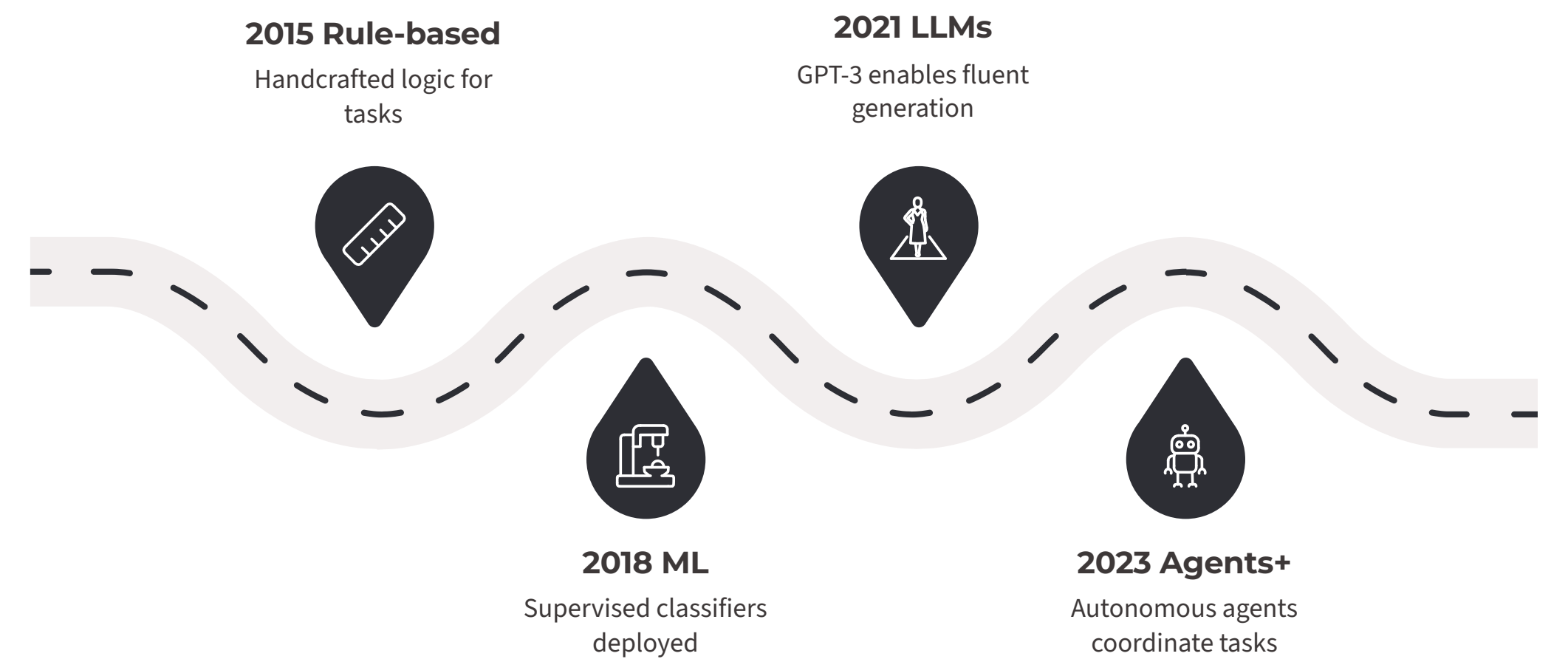
- You accept OCR is not 100% accurate (especially handwriting)
- You will not upload copyrighted/illegal content
- Processing is in-memory only
- Do not reverse-engineer or attack the service

What We Promise

- Process in memory only — never stored
- Display confidence so you can verify
- Maintain service & fix vulnerabilities responsibly
- Attribute open-source dependencies

 Like a photocopying shop — we copy your document, we don't keep a copy, and we're not liable for what you copied.

We're Building for the Agentic Era — And Building Our Careers



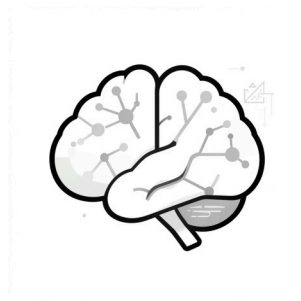
Technical

AI-Era CV (OpenCV), OCR engineering (Tesseract/EasyOCR), multi-PSM strategy, cloud deployment, Python OOP & modular code



Professional

Documentation, team collaboration, Git workflows, reproducibility, peer review



Agentic

Agent architectures, memory & learning systems, human-in-the-loop design, explainable AI

✔ Every major tech company is hiring AI Agent Engineers in 2025–2026. This project is our portfolio entry point.

Building Remotely · Building Responsibly · Building Green

Virtual Collaboration

- GitHub = full audit trail of every commit
- Asynchronous work — no need to be co-located
- Issues + PR comments = remote code review
- Commit history = transparent record of contributions
- Modular task division: preprocessing / OCR / agents / UI

Green Computing Choices

Choice	Eco Impact
opencv-python-headless	Less RAM → less energy
Streamlit Cloud (serverless)	Resources used only when active
Session-only processing	No idle DB load
Adaptive preprocessing	Heavy work only when needed (quality <65)
On-demand deployment	No always-on backend

- ❏ Compared to an always-on GPU server, our serverless on-demand architecture uses **~100× less energy**. Efficient code is green code.